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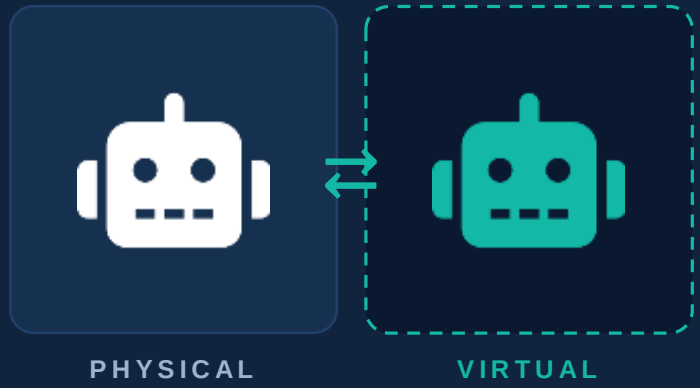
ROBOTICS TRAINEE PROGRAMME · THIS WEEK

Digital Twins for Robotics



live data · both ways

Build a living virtual copy of a real machine — and learn why every serious robotics team on earth now tests in software before touching hardware.

Lecture pack · Intermediate track · Includes simulation + physical weekend practicals

What Is a Digital Twin?

A digital twin is a virtual model of a physical object or system, continuously updated with real data from that object, and able to influence it in return.

Three ingredients — remove any one and it is no longer a twin:

1

The physical thing

A motor, a robot, a generator, a whole factory — something real that exists and behaves.

2

The virtual model

A software representation: geometry, physics, behaviour — detailed enough to answer real questions.

3

The live connection

Sensors stream reality into the model; decisions flow back. This link is what separates a twin from a mere simulation.

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Model vs Shadow vs Twin: The Data-Flow Test

Industry abuses the word “twin.” Classify any system by asking: which way does the data flow automatically?

Digital MODEL	None — built by hand, updated by hand	*A CAD file of your robot; a Gazebo world you edited manually*
Digital SHADOW	Physical → virtual (one way, automatic)	*A dashboard that live-plots your robot's battery and pose*
Digital TWIN	Physical ⇄ virtual (both ways, automatic)	*The model detects motor overheating and commands the real robot to slow down*

Exam question guaranteed: give one Nigerian example of each. Start thinking now.

Anatomy of a Twin: The Loop

1 · Physical asset

robot, motor, plant

2 · Sensors

IMU, encoders, temp, LiDAR

3 · Data pipeline

serial, MQTT, cloud

(wraps to stage 4, bottom row → read left to right)

4 · Virtual model

physics + state

5 · Analytics / AI

predict, detect, optimise

6 · Actuation

commands back to hardware

Stage 6 feeds back into Stage 1 — the loop never stops. A twin is a control loop wrapped around the whole system, not a picture of it.

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How Deep Should the Twin Go? Fidelity Levels

Fidelity is a budget decision. Match the model depth to the question you need answered — no deeper.

L1 · Geometric	Shape and layout only	*“Will the robot fit through that corridor?”*
L2 · Kinematic	Motion without forces	*“Where is the arm's gripper when joint 2 is at 40°?”*
L3 · Physics / dynamic	Forces, friction, inertia, torque	*“Will this payload tip the robot on a ramp?”*
L4 · Behavioural + data-driven	Learned models, wear, failure statistics	*“Which motor will fail first, and when?”*

This week's practicals live at L1–L2. RAIN's drone work lives at L3. Predictive maintenance contracts live at L4.

Why Robotics Cannot Live Without Twins

Crash in software, not steel

A bad path in simulation costs nothing. The same bug on a physical robot costs a chassis, a motor driver — or a teammate's ankle.

Test 10,000× faster

Simulated robots run faster than real time and in parallel. One weekend of simulation = months of physical trials.

Train AI safely

Reinforcement learning needs millions of trials. You cannot do millions of trials on one physical robot — you can in its twin.

Predict before failure

A twin fed with vibration and temperature data flags a failing bearing weeks early. Downtime becomes a scheduled event.

Remember last week's rule: hardware is the last step, never the first.

Where Twins Are Working Today

Manufacturing

Factory lines tuned virtually before retooling; robots re-programmed offline while production keeps running.

Power & energy

Grid and turbine twins predict faults. Imagine a twin of a Nigerian feeder line flagging overloads before load-shedding.

Smart cities

Traffic twins test interventions virtually. A Lagos BRT-corridor twin could trial signal timings without touching a single junction.

Healthcare

Patient-specific heart and organ models let surgeons rehearse procedures before the first incision.

Agriculture

Farm twins fuse soil sensors, weather and drone imagery — irrigation and fertiliser decisions tested in software first.

Robotics & drones

Every serious lab flies the mission in the twin first. This is exactly how RAIN prototypes drone behaviour.

The Twin-Builder's Toolbox

Sense	Arduino / ESP32, IMUs, encoders, ultrasonic, DHT22, current sensors	*the nervous system*
Transport	Serial/USB for the bench; MQTT + Wi-Fi for the field; databases for history	*the bloodstream*
Model & simulate	Python (NumPy, Matplotlib), Webots, Gazebo, PyBullet, URDF robot descriptions	*the imagination*
Decide & display	Dashboards (Matplotlib live plots, Grafana, web apps), anomaly detection, ML models	*the brain and face*

Everything on this slide is already in the RAIN lab or free to install. No excuses.

⚖ The Sim-to-Real Gap: Where Twins Lie

A twin is only as honest as its calibration. Reality always disagrees with the model somewhere:

Friction & slip
Your simulated wheels never slip. Your real wheels slip on RAIN's tiled floor every single run.

Sensor noise
Simulated ultrasonic returns 42.0 cm. The real HC-SR04 returns 41, 44, 39, 213(!), 42…

Latency
In simulation, commands act instantly. In reality: serial delay + motor spin-up + battery sag.

Battery & wear
Motors weaken as voltage drops. A twin that ignores battery state drifts within minutes.

💡 **The professional habit — calibrate:** measure the real system, feed measurements back into the model, repeat until twin and reality agree within tolerance. That loop IS the job.

Case Study: Twinning This Week's Grid Runner

Everything from the pathfinding lecture becomes a digital twin the moment we close the loop:

1. **The occupancy grid** is the twin's world model — a live map of the arena.
2. **The A* planner** is the twin's analytics layer — it reasons about the world model.
3. **The overhead photo** is the sensor feed — physical → virtual, updating the map.
4. **The command string (F, L, R)** is actuation — virtual → physical, closing the loop.
5. **Re-photograph + re-plan when a box moves** and congratulations: you have built a genuine two-way digital twin.

One project, two lectures, one loop. This is deliberate.

A Minimal Twin in 30 Lines

Arduino streams reality; Python mirrors it live. This skeleton is your weekend starting point.

Arduino — the physical side

```
// HC-SR04 -> serial stream
long readDistanceCm() {
  digitalWrite(TRIG, LOW);  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
  return pulseIn(ECHO, HIGH) / 58;
}
void loop() {
  Serial.println(readDistanceCm());
  delay(100);   // 10 Hz
}
```

Python — the virtual side

```
import serial, collections
import matplotlib.pyplot as plt

ser  = serial.Serial('COM5', 9600)
hist = collections.deque(maxlen=100)

plt.ion()
while True:
    d = float(ser.readline())
    hist.append(d)          # twin state
    plt.cla()
    plt.plot(hist); plt.ylim(0, 200)
    plt.title(f'Twin: {d:.0f} cm')
    plt.pause(0.01)
```

Add a rule — "if d < 15, send STOP back over serial" — and the shadow becomes a twin.

Exercise Ladder — Climb During the Week

1 Classify it BEGINNER

For 6 systems (Google Maps traffic, a CAD drawing, a CBT exam dashboard, a thermostat, a flight simulator, RAIN's LMS analytics), label each: model, shadow, or twin — and defend your answer in one sentence.

2 Spec a twin on paper BEGINNER

Pick any machine in the RAIN lab. List: 3 sensors you would attach, the data each produces, one decision the twin could automate. One page max.

3 Kinematic twin INTERMEDIATE

Write a Python class DiffDriveTwin(x, y, theta) that updates pose from wheel speeds (unicycle model). Animate it driving a square with matplotlib.

4 Noise modelling INTERMEDIATE

Log 200 real HC-SR04 readings of a fixed target. Plot the histogram, compute mean and std-dev, then make your simulated sensor produce the same distribution.

5 Predictive rule ADVANCED

Stream motor current while loading the shaft by hand. Design a rule (threshold or moving average) that flags “abnormal load” — your first predictive-maintenance twin.

Exercises 1–3 due before Friday's lab. 4–5 earn bonus marks.

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Weekend Practical A — Simulation Mission

SOLO OR PAIRS | PYTHON + MATPLOTLIB (WEBOTS OPTIONAL) | SUBMIT: CODE + 60-SEC SCREEN RECORDING

"The Ghost Robot" — a kinematic twin that replays your A mission*

1. Build DiffDriveTwin: state (x, y, θ), method step(v_left, v_right, dt) using the unicycle/differential-drive equations.
2. Feed it the F/L/R command string from your pathfinding weekend task. The twin should trace the same route through the same grid — draw both.
3. Inject realism: add Gaussian noise to wheel speeds (σ = 5%). Run 20 trials and plot all 20 ghost trajectories over the ideal path.
4. Measure the drift: report mean final-position error across the 20 runs. This number is your sim-to-real gap, quantified.
5. Stretch: implement simple closed-loop correction (re-aim at the next waypoint each step) and show the error shrink.

If you have Webots installed: replicate steps 1–2 with the e-puck robot in a matching arena for extra credit.

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Weekend Practical B — Physical Build

TEAMS OF 3 | ARDUINO + HC-SR04 + SERVO (LAB KITS) | LAPTOP RUNNING THE PYTHON TWIN

"The Room Twin" — a live radar map of a real corner of the lab

1. Mount the HC-SR04 on a servo. Sweep 0–180° in 5° steps; at each angle print "angle,distance" over serial.
2. Python side: read the stream and convert to Cartesian points — $x = d \cdot \cos(\theta)$, $y = d \cdot \sin(\theta)$. Plot a live polar/scatter map.
3. Place 2–3 boxes in front of the rig. Your twin's map should show them as point clusters. Move a box — the map must follow within one sweep.
4. Close the loop: if any object enters the 20 cm "keep-out zone," Python sends 'A' back over serial and the Arduino sounds a buzzer / LED.
5. Calibrate honestly: measure one box with a tape rule vs your map. Report the error in cm — every good twin ships with an accuracy figure.

Deliverable: 2-minute video — rig, live map reacting to a moving box, buzzer firing. Demo day: Monday 9 AM.

Key Takeaways

- A digital twin = physical asset + virtual model + a live two-way data link. All three, always.
- Twins let robotics teams crash in software, test thousands of times faster, and train AI safely.
- The data-flow test separates model (no auto flow), shadow (one way) and twin (both ways).
- The sim-to-real gap is permanent — calibration against real measurements is the core professional skill.
- Fidelity is a budget: model only as deep as the question you need answered.
- Your weekend projects ARE twins: the sweep map mirrors reality; the STOP command closes the loop.

Next week we join the two threads: the Grid Runner drives inside its own twin, and the twin re-plans with A when the world changes.*